Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **distributed systems**, **database pipelines**, and **Java/C++** projects; optimizing **queries**, scheduling, reliability, and storage-backed workloads.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reduced dashboard latency from 1.2s to 350ms through **query optimization**, indexing, and normalized Azure SQL schemas, designing row-level isolation for a multi-tenant data platform supporting more than 500 users.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant distributed execution** with FastAPI workers, explicit state transitions, and failure isolation, preventing cascading failures while maintaining reliable background processing.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Enabled **concurrent batch execution** of 1K+ files using Python multithreading and secure ingestion, integrating AWS RDS and REST APIs while preventing long-running grading workloads from blocking instructor-facing application workflows.
- Sustained 10K+ daily grading requests for 150+ instructors through a **production Flask backend**, CI/CD releases, and structured persistence, reducing feedback turnaround by 60% while maintaining stable responses during peak loads.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Improved long-horizon trajectory planning by 18% in simulation using a Decision Transformer, building **distributed data pipelines** that processed 100K+ trajectory logs for offline reinforcement-learning training and evaluation workloads.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder**, integrating HD-map geometry and ego-agent history with multi-head attention while benchmarking architectural alternatives across repeatable simulation experiments.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **indexing, caching, and pagination**, applying object-oriented patterns to cross-platform integration modules while reducing payloads and optimizing data access for low-bandwidth client environments.
- Increased ETL efficiency by 68% through **AWS Glue and S3** transformations; automated executable builds and 20+ unit/integration suites across Windows, Linux, and macOS using Jenkins, PyTest, and Mockito.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*

- Reduced centralized-control and quantum-risk exposure by 25% in simulations through a **distributed consensus algorithm**, evaluating witness selection and lattice-based cryptography while balancing security, fairness, and protocol performance trade-offs.
- Published peer-reviewed systems research after integrating 5+ **cryptographic primitives** into a quantum-resistant blockchain design, presenting experimental methodology and results at IISc Bengaluru while documenting architecture, algorithms, and benchmark findings.

Skills

Languages: Java, C++, Python, SQL, JavaScript, TypeScript, Bash
Data Systems: Distributed Systems, Database Systems, Storage Systems, Query Optimization, Indexing, Caching, Data Modeling, Transaction Processing, Resource Management
Databases & Processing: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, Redis, Snowflake, AWS Glue, ETL Pipelines
Backend & Infrastructure: FastAPI, Flask, REST APIs, Microservices, Docker, Kubernetes, AWS S3, AWS RDS, Linux
Engineering: Object-Oriented Programming, Data Structures, Algorithms, Multithreading, Performance Optimization, Fault Isolation, Unit Testing, Integration Testing, CI/CD

Projects

Vectorized Query Pipeline - Design Concept

- Proposed a **Java query pipeline** with logical planning, cost-based rule selection, partition-aware scheduling, vectorized batches, and indexed storage, outlining benchmarks for throughput, latency, skew, and recovery without implementation claims.

CPU-scheduling Simulator | Live

- Implemented **C++ scheduling algorithms** including FCFS, SJF, SRT, Round Robin, priority, and MLQ, validating preemptive and non-preemptive behavior with interactive visualization and wait-time calculations across 1K+ simulated process combinations.

Chat2SQL | Code

- Unified **10+ database systems** in a Java and TypeScript workspace extending Chat2DB, integrating schema exploration, query generation, structure synchronization, and Docker-based development across Windows, macOS, and Linux environments.